

How to flash the binaries of the ESP32 edition of TCode controller

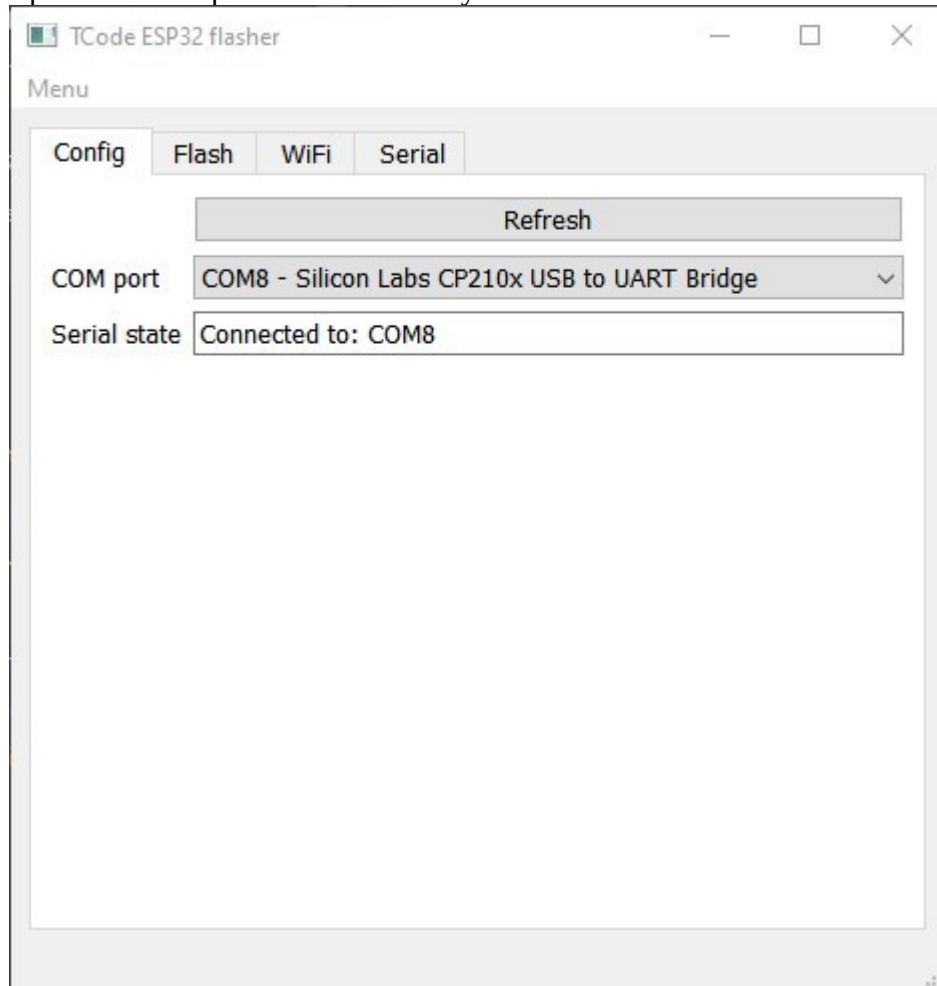
Extract the zip archive.

(Linux/Mac users [click here](#) See command `example.txt` for the command to run)

(Windows users: this may work in other OS' with an API layer (Wine/Parallels) I've not tested.

Run “flash.exe”

Select your com port in the dropdown if not already selected. You should see connected.



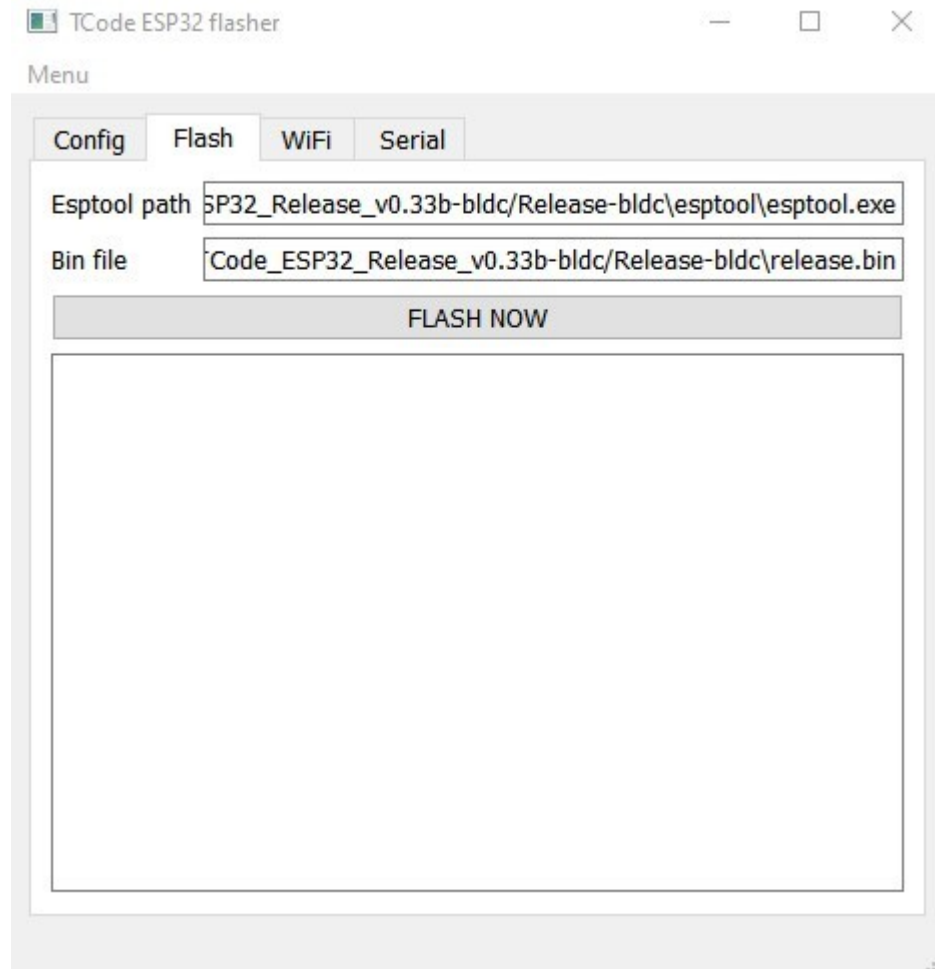
If your ESP32 isn't recognized as a COM port you may need to install the drivers for your USB chip.

For micro USB Devkit <https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers>

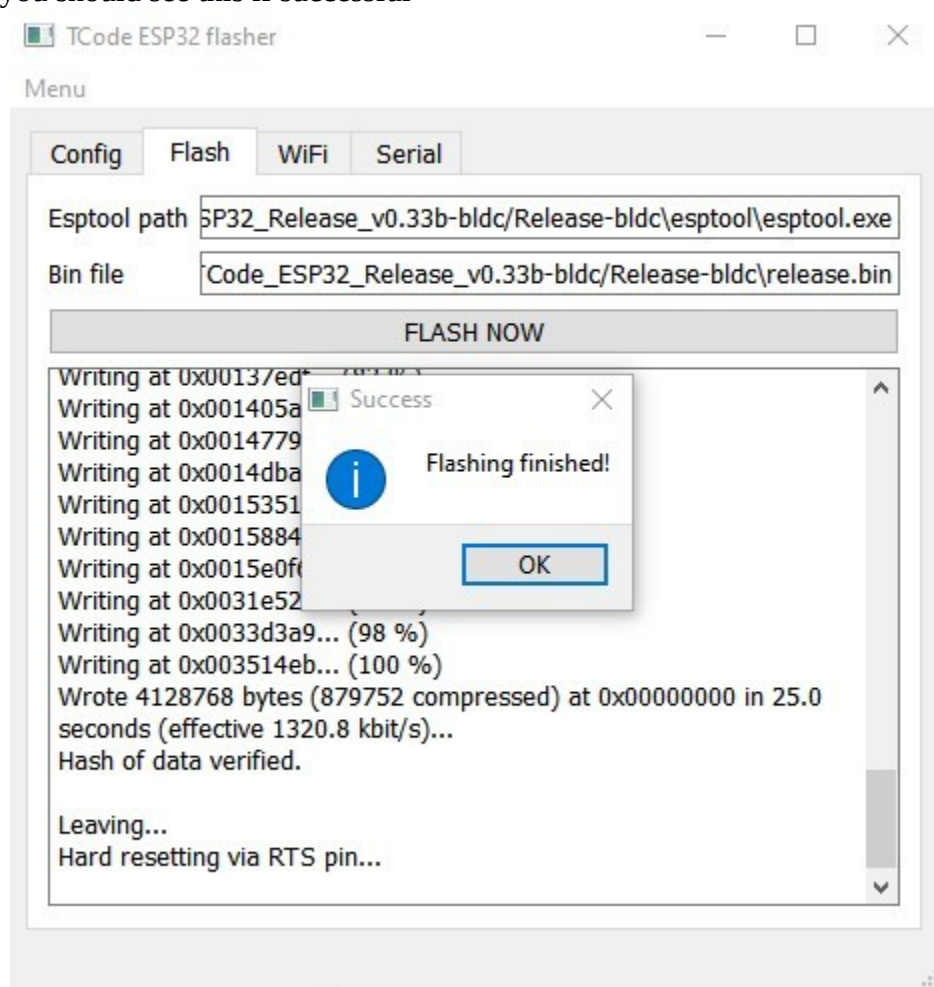
For USB-C Devkit (CH340) <https://learn.sparkfun.com/tutorials/how-to-install-ch340-drivers/all>

Select the Flash tab (most should leave everything at default on this tab) If using the S3 Zero of another supported board, Select it in the module drop down (Not pictured) and click “FLASH NOW”

Note: If you get a error that mentions the boot mode, hold the boot button in the Devkit until the dots stop.



After flashing you should see this if successful



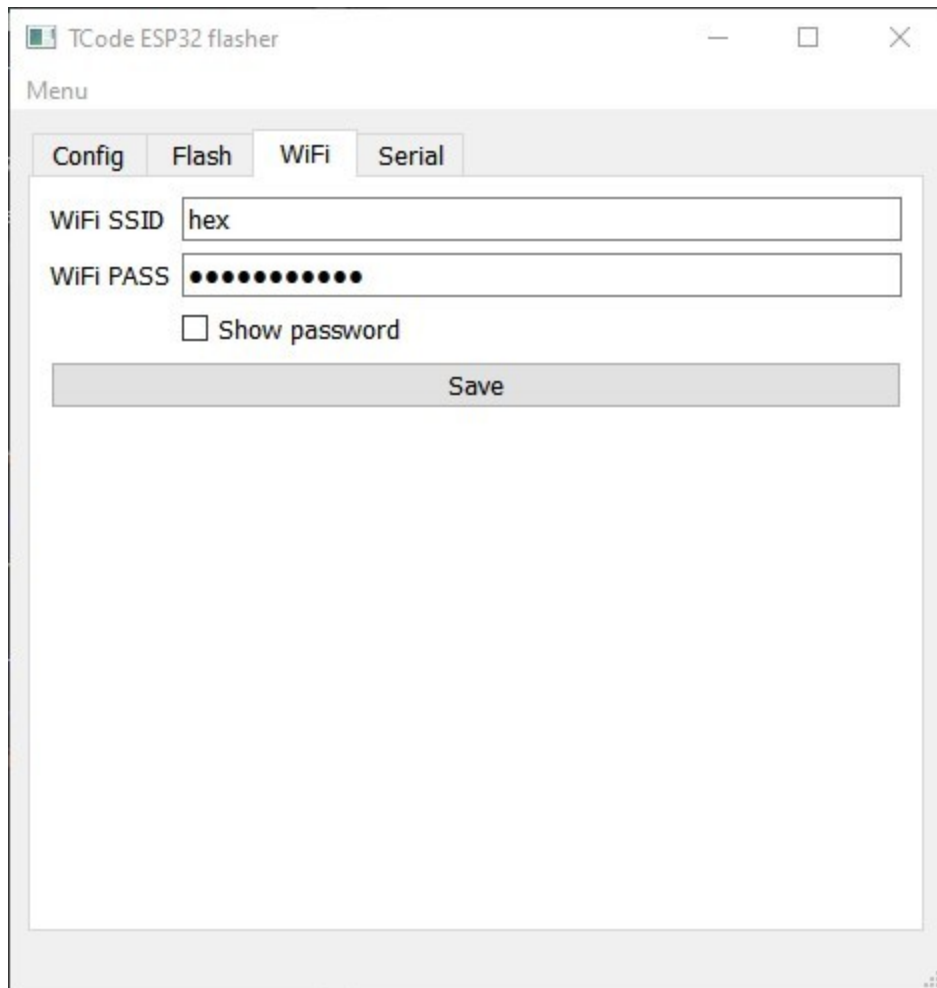
Now that your image is flashed time to configure the wifi if you wish to do so
IMPORTANT! The ESP32 S series (Devkit v1) is ONLY compatible with 2.4ghz WiFi

You can either configure Wifi with the flasher, AP mode or Serial monitor.
Skip to the chosen section below.

Flasher configuration:

Select the WiFi tab and enter your 2.4ghz SSID and password

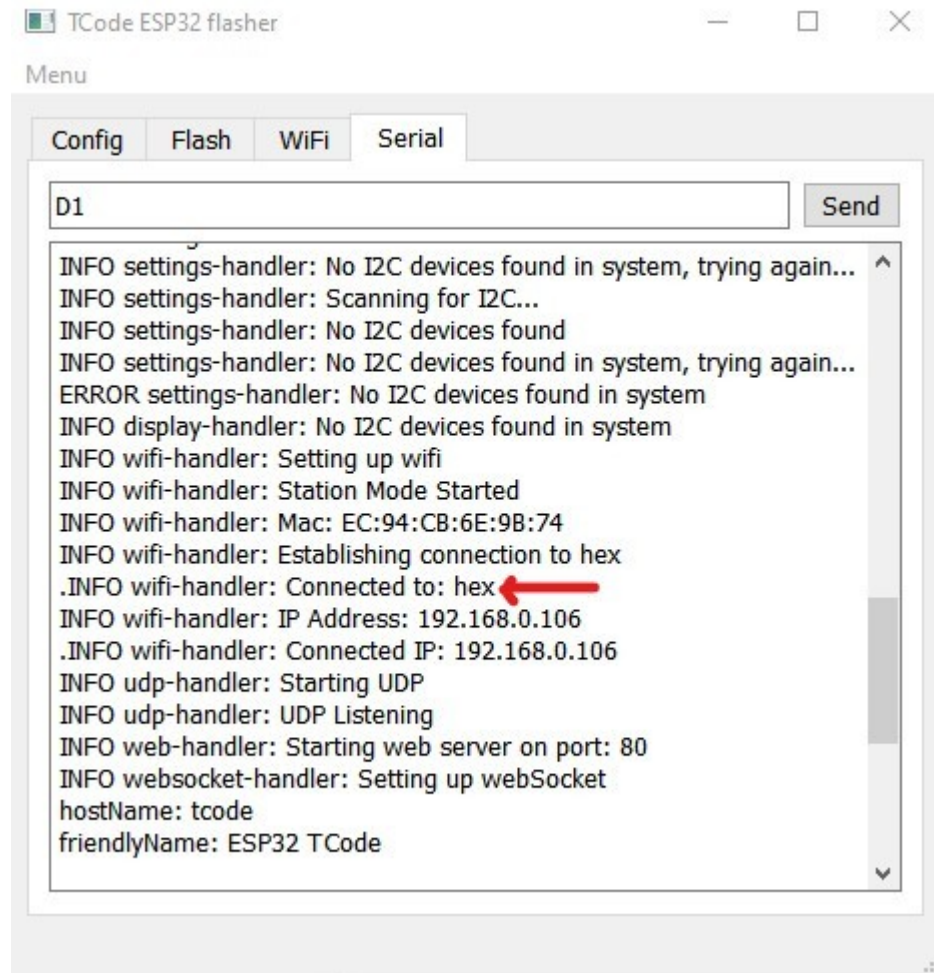
Click Save



The image shows a software window titled "TCode ESP32 flasher". It has a "Menu" bar with four tabs: "Config", "Flash", "WiFi", and "Serial". The "WiFi" tab is currently selected. Below the tabs, there are two input fields: "WiFi SSID" containing the text "hex" and "WiFi PASS" containing ten black dots. Below these fields is a checkbox labeled "Show password" which is currently unchecked. At the bottom of the form area is a large grey button labeled "Save".

On success, the app will switch to the Serial tab to view the output.

Here you can validate the ESP32 connects to your network.

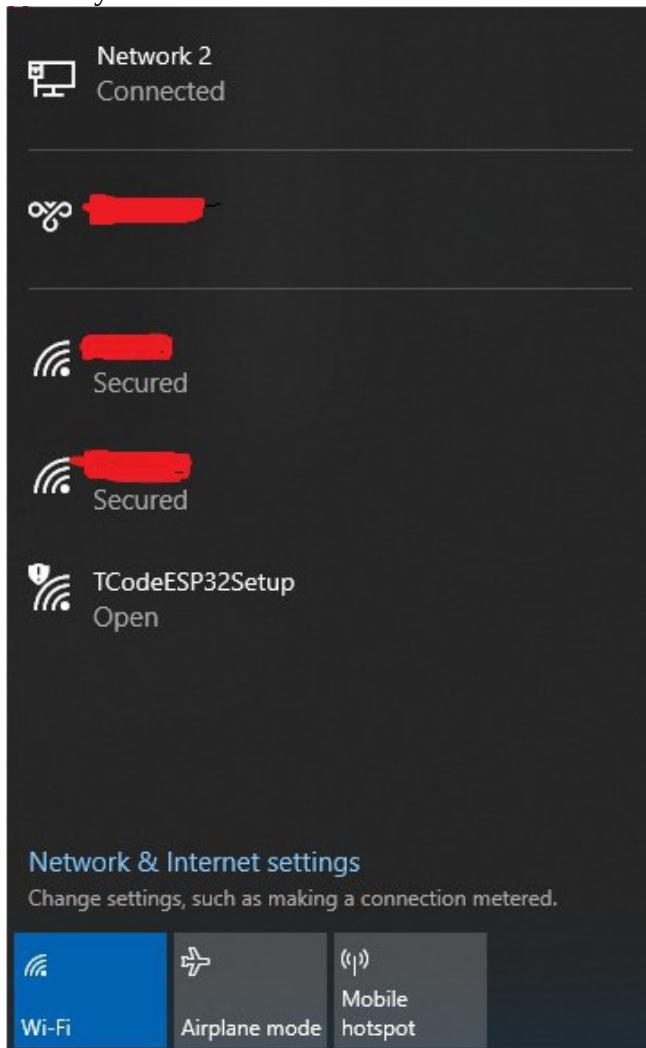


You can also enter other Tcode commands here to test your device is working correct.

IMPORTANT! On BLDC devices, You must set the DeviceType and EncoderType via serial or web page before the device will work.

AP configuration:

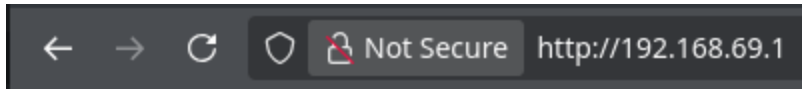
Check your available wifi networks



Connect to TcodeESP32Setup (leave connect automatically **unchecked**)

Password is: **tcode_6969**

Once connected (It can be slow be patient. About 30 secs or so)



Open your internet browser and navigate to 192.168.69.1

Enter your wifi ssid and password and change the network info if required.

Note: If you're NOT connecting to an AP/Router SSID, you may want to set a different AP Mode SSID/Password for privacy/security. Connecting the device to a router will disable AP Mode unless the network can't be reached for some reason.

You can also change other settings such as board type and device type.

Board type will change the pinout for the board

Device type will change the setup page and the way the code handles the signals.

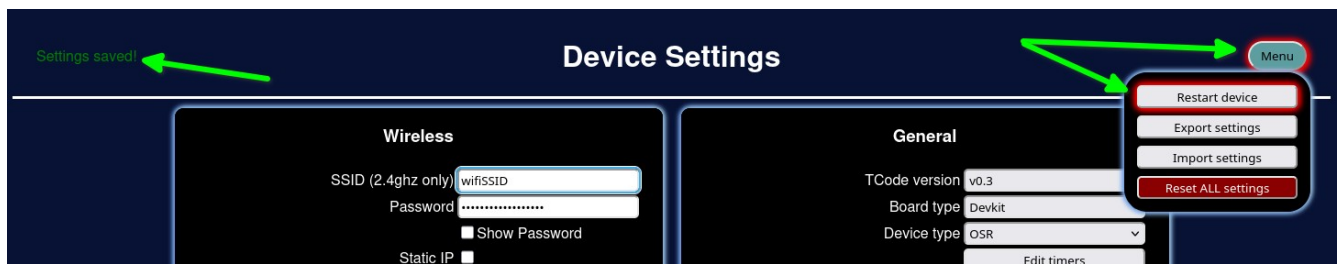
You can change other settings or you can wait until reboot happens.

BLDC based devices have an encoder type field

A screenshot of the 'Device Settings' web interface. The page has a dark blue header with the title 'Device Settings' and a 'Menu' button. The main content is divided into four panels: 'Networking', 'General', 'Other', and 'Sleeve temperature'. The 'Networking' panel includes fields for SSID, Password, Static IP, UDP port, Web port, Host name, MDNS enabled, Friendly name, AP mode SSID, AP mode password, AP Mode IP, AP Mode subnet, AP Mode gateway, AP Mode hidden, AP Mode channel, and BLE enabled. The 'General' panel includes TCode version, Board type, Device type, Encoder, Motor settings, Use hall effect, PID proportional const, Low pass filter, Servo resolution, Edit timers, a Servo table, and PWM available on. The 'Other' panel includes Inverse T-Valve, T-Valve 90 degree servo, Inverse main servos, Inverse Pitch, and Inverse twist. The 'Sleeve temperature' panel includes Enabled, Sleeve temp sensor pin, Heater pin, Heater timer channel, and Target Temperature. The Servo table has columns for Servo, PIN, ZERO (μ), and Timer Channel, with rows for Valve, Twist, and Aux.

Servo	PIN	ZERO (μ)	Timer Channel
Valve	12	1500	High 3 CH6
Twist	13	1500	High 2 CH4
Aux	17	1500	High 2 CH5

The settings will auto save. Wait for the Settings saved text to appear and the Menu/Restart device buttons will flash..



Click restart device OR unplug and re-plug the usb powering the ESP32.

Your device should reboot and connect to the network if you entered a valid SSID/Password.

You can verify this by using serial monitor:

```

PROBLEMS  OUTPUT  TERMINAL  GITLENS  DEBUG CONSOLE

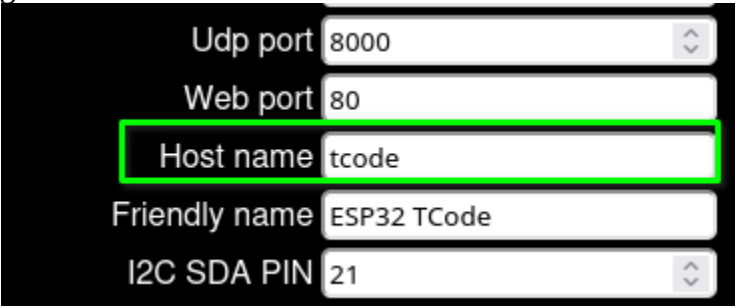
SPI_FAST_FLTA
INFO: ESP32 Chip model = ESP32-D0WDQ6 Rev 1
INFO: This chip has 2 cores
INFO: Chip ID: 15859308
INFO: Read Settings: /userSettings.json
INFO: Last reset reason: Reset due to power-on event
INFO: Version: ESP32 v0.251b
INFO: Setting up wifi
INFO: Station Mode Started
INFO: Mac: 24:62:AB:F1:FE:6C
INFO: Establishing connection to hex
WARNING: Disconnected from station, attempting reconnection
INFO: Reason: 0
INFO: Unknown reason 0
INFO: Connected to [REDACTED]
INFO: IP Address: 192.168.0.95
INFO: Connected IP: 192.168.0.95
INFO: Starting UDP
INFO: UDP Listening
INFO: Starting web server on port: 80
INFO: Setting up webSocket
hostName: tcode
friendlyName: ESP32 TCode
ESP32 v0.251b
TCode v0.3
Ready!
[

```

or checking your router dhcp for the ESP32.

10	TCodeESP32		192.168.0.145
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You should now be able to access the configuration page from [or](#) what ever you type into the Host name field on the configuration.



Udp port 8000

Web port 80

Host name tcode

Friendly name ESP32 TCode

I2C SDA PIN 21

Once you have this IP address you can get into your machine settings via the web browser.

From here you can view the [default pin out](#) and change them if you know what you are doing. -1 means disabled. You cant disable the main servos for the selected Device type

You may want to set up things like servo frequency in the [timer setup](#)

Or calibrate your servo [zero positions](#).

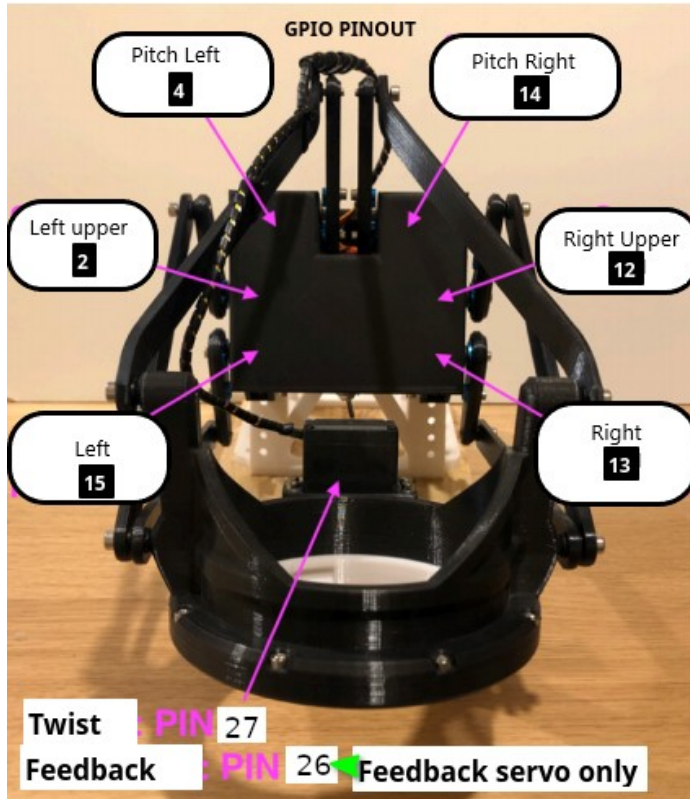


Main servos max range (°C) 180

Edit timers

Servo	PIN	ZERO (μ)	Timer Channel
Right	13	1500	High 0 CH0
Left	15	1500	High 0 CH1
Right upper	12	1500	High 1 CH2
Left upper	2	1500	High 1 CH3
Pitch	4	1500	High 2 CH4
Pitch right	14	1500	High 2 CH5
Valve	25	1500	High 3 CH7
Twist	27	1500	High 3 CH6
Aux	-1	1500	None

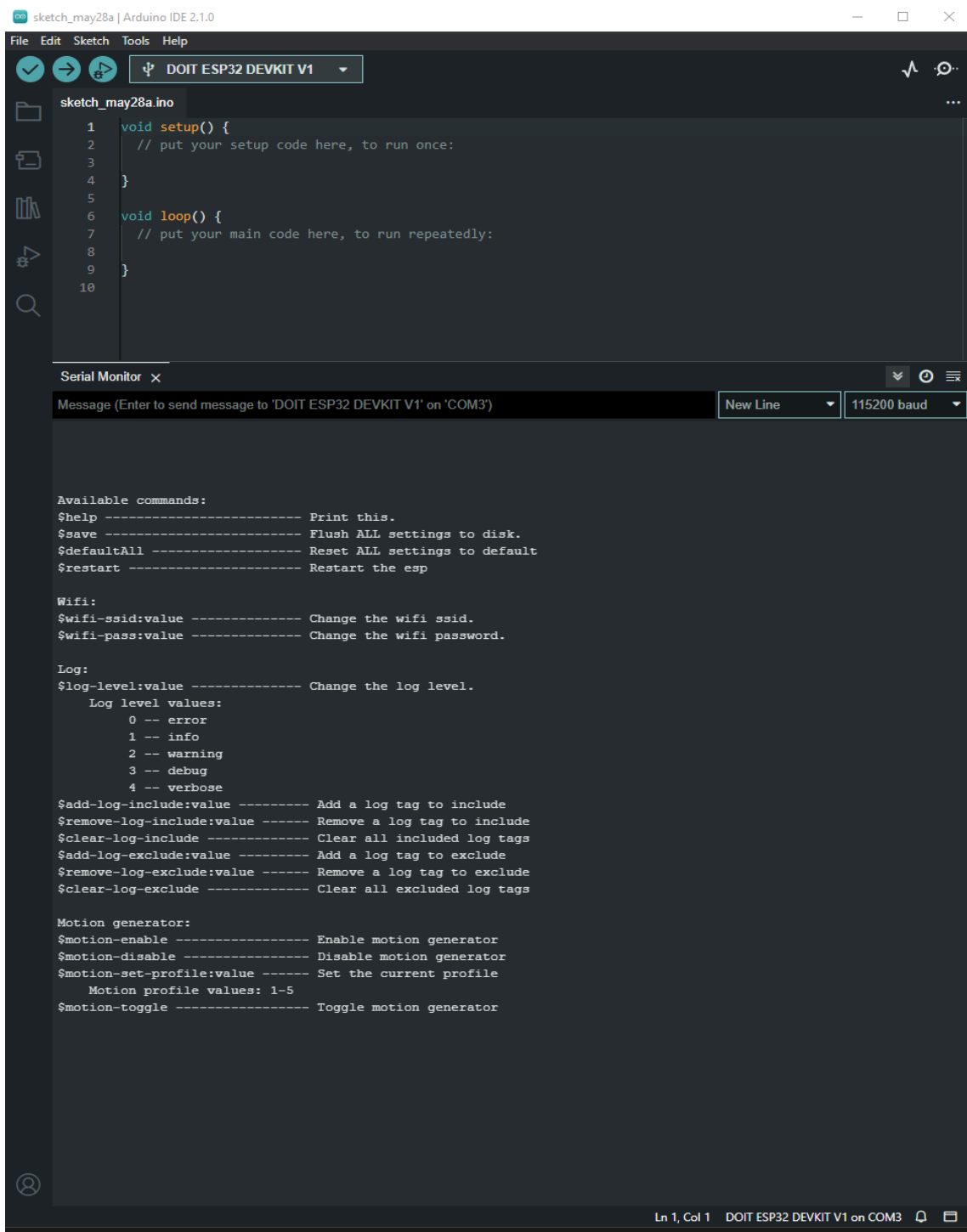
Here is the default SR6 pinout for reference:



Serial monitor configuration

If you can't connect to the APMode method above for some reason you can configure most device settings over serial communications including the wifi router login settings.

Connect to the esp32 via Serial monitor in Arduino ide or your app of choice



```
sketch_may28a.ino
1 void setup() {
2   // put your setup code here, to run once:
3
4 }
5
6 void loop() {
7   // put your main code here, to run repeatedly:
8
9 }
10
```

Serial Monitor x

Message (Enter to send message to 'DOIT ESP32 DEVKIT V1' on 'COM3') New Line 115200 baud

Available commands:

\$help ----- Print this.

\$save ----- Flush ALL settings to disk.

\$defaultAll ----- Reset ALL settings to default

\$restart ----- Restart the esp

Wifi:

\$wifi-ssid: value ----- Change the wifi ssid.

\$wifi-pass: value ----- Change the wifi password.

Log:

\$log-level: value ----- Change the log level.

Log level values:

0 -- error

1 -- info

2 -- warning

3 -- debug

4 -- verbose

\$add-log-include: value ----- Add a log tag to include

\$remove-log-include: value ----- Remove a log tag to include

\$clear-log-include ----- Clear all included log tags

\$add-log-exclude: value ----- Add a log tag to exclude

\$remove-log-exclude: value ----- Remove a log tag to exclude

\$clear-log-exclude ----- Clear all excluded log tags

Motion generator:

\$motion-enable ----- Enable motion generator

\$motion-disable ----- Disable motion generator

\$motion-set-profile: value ----- Set the current profile

Motion profile values: 1-5

\$motion-toggle ----- Toggle motion generator

Ln 1, Col 1 DOIT ESP32 DEVKIT V1 on COM3

Enter the command `#help` to see a list of available commands.

IMPORTANT! On BLDC devices, You must set the DeviceType and EncoderType via serial or web page before the device will work.

Conclusion

Now wifi is connected you can get to the web page via either <http://tcode.local> or `http://<ipaddress>` and configure the rest this firmware.

Enjoy!

